ATHÉNÉE INVESTMENT × EPFL

Lazarus Alpha AMC

Diversification Benefits & Drawdown Mitigation in Multi-Manager Portfolios

Confidential Institutional Memorandum & Strategy Whitepaper
Target Audience: Investment Committees, Family Offices, and Strategic Partners

Abstract:
Traditional equity and long-biased alternative allocations exhibit structural vulnerability to macroeconomic drawdowns and tail risk clustering. This paper presents the mathematical, empirical, and operational framework for an institutional Multi-Manager Market-Neutral Strategy. By assembling a diversified portfolio of $N \geq 10$ uncorrelated, capacity-constrained quantitative and niche relative-value hedge funds, the portfolio exploits the non-linear $1/\sqrt{N}$ volatility compression theorem. Combining rigorous empirical backtesting across the J.P. Morgan hedge fund universe (2020–2026) with institutional sourcing from Tier-1 Prime Brokerage networks and ex-pod shop spin-offs, we demonstrate that a disciplined allocation framework compresses 10-year Maximum Drawdown (MDD) by a factor of 4.0x–6.0x while delivering double-digit net absolute returns with zero structural market beta.

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1. Executive Summary & Core Thesis

Institutional allocators face a persistent dilemma: public equities present high volatility (∼18%) and deep multi-year drawdowns (30–50%), while traditional alternative portfolios frequently suffer from hidden equity beta and correlation spikes during liquidity crises.

Our proposed product is an institutional Multi-Manager Market-Neutral Fund designed to solve this dilemma through structural diversification across high-conviction, orthogonal alpha streams.

Key Strategy Pillars

- **Orthogonal Alpha Generation:** Direct allocation to $N = 10$ to 14 capacity-constrained, market-neutral hedge fund strategies with **proven non-correlation** ($|\beta| < 0.1$) to broader market factors and each other.
- **Mathematical Volatility Crushing:** Exploitation of portfolio variance dampening where portfolio volatility scales asymptotically towards $\sigma/\sqrt{N}$, compressing portfolio volatility from 18% down to 6.0% – 7.5%.
- **Drastic Drawdown Compression:** Monte Carlo modeling confirms a theoretical 5.5x–6.0x reduction in 10-year Maximum Drawdown (MDD), going from S&P 500 MDD of 30% – 40% to < 10%.
- **Elite Institutional Sourcing:** Leveraging top-tier Prime Brokerage (PB) Capital Introduction desks (JPM, GS, BNP, Citi, Barclays, UBS), spin-offs from major pod shops (e.g., Millennium, Citadel, Point72) and bank ex - prop desks with strict built-in drawdown budgets (5–7%).
- **Statistical Edge Verification:** Prioritizing high-turnover systematic strategies where statistical significance (t-stat > 2.0) distinguishes genuine alpha from random variance.

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2. The Mathematics of Orthogonal Diversification

2.1 Theoretical Volatility Scaling ($1/\sqrt{N}$ Law)

Consider a portfolio of N equally weighted assets, each exhibiting annualized volatility $\sigma_i = \sigma$ and uniform pairwise correlation $\rho_{ij} = \rho$ for all $i \neq j$. The portfolio variance σ_p^2 is defined analytically as:

$$\sigma_p^2 = \frac{1}{N^2}\sum_{i=1}^{N}\sigma_i^2 + \frac{1}{N^2}\sum_{i=1}^{N}\sum_{j\neq i}^{N}\sigma_i\sigma_j\rho \tag{1}$$

Assuming identical volatilities $\sigma_i = \sigma$, this simplifies to:

$$\sigma_p = \sigma\sqrt{\frac{1+(N-1)\rho}{N}} = \sigma\sqrt{\frac{1-\rho}{N}+\rho} \tag{2}$$

Under the condition of true orthogonality ($\rho \to 0$):

$$\lim_{\rho\to 0}\sigma_p = \frac{\sigma}{\sqrt{N}} \tag{3}$$

For a single standalone strategy with $\sigma = 18.0\%$, diversifying across $N = 10$ orthogonal assets yields:

$$\sigma_p = \frac{18.0\%}{\sqrt{10}} \approx 5.69\%$$

Even in the presence of modest residual baseline correlation ($\rho = 0.10$):

$$\sigma_p = 18.0\% \times \sqrt{\frac{1 + 9(0.10)}{10}} = 18.0\% \times \sqrt{0.19} \approx 7.85\%$$

2.2 Monte Carlo Drawdown Compression Analysis

To analyze the path-dependent behavior of wealth trajectories over a 10-year horizon ($T = 120$ months), we simulated 10,000 Monte Carlo paths comparing:

1. **Standalone Asset (S&P 500 Proxy):** $\mu = 10.0\%$, $\sigma = 18.0\%$.
2. **Synthetic Multi-Manager Portfolio ($N = 10$, $\rho = 0$):** 10 independent copies of $\mu = 10.0\%, \sigma = 18.0\%$ with monthly rebalancing.
3. **Conservative Multi-Manager Portfolio ($N = 10$, $\rho = 0.10$):** Incorporating realistic correlation friction.

Portfolio Configuration	Ann. Return	Ann. Vol	10-Yr Median MDD	10-Yr 25th Pct MDD
Single Asset (S&P 500 Proxy)	10.0%	18.0%	30.2%	40.5%
Orthogonal Portfolio ($N = 10, \rho = 0.00$)	10.0%	**5.7%**	**5.2%**	**7.1%**
Realistic Portfolio ($N = 10, \rho = 0.10$)	10.0%	**7.8%**	**8.1%**	**11.0%**
High-Alpha Manager Sleeve ($N = 10, \rho = 0.05$)	**24.0%**	**6.8%**	**6.4%**	**8.9%**

Table 1: Monte Carlo 10-Year Horizon Simulation Statistics (10,000 Iterations).

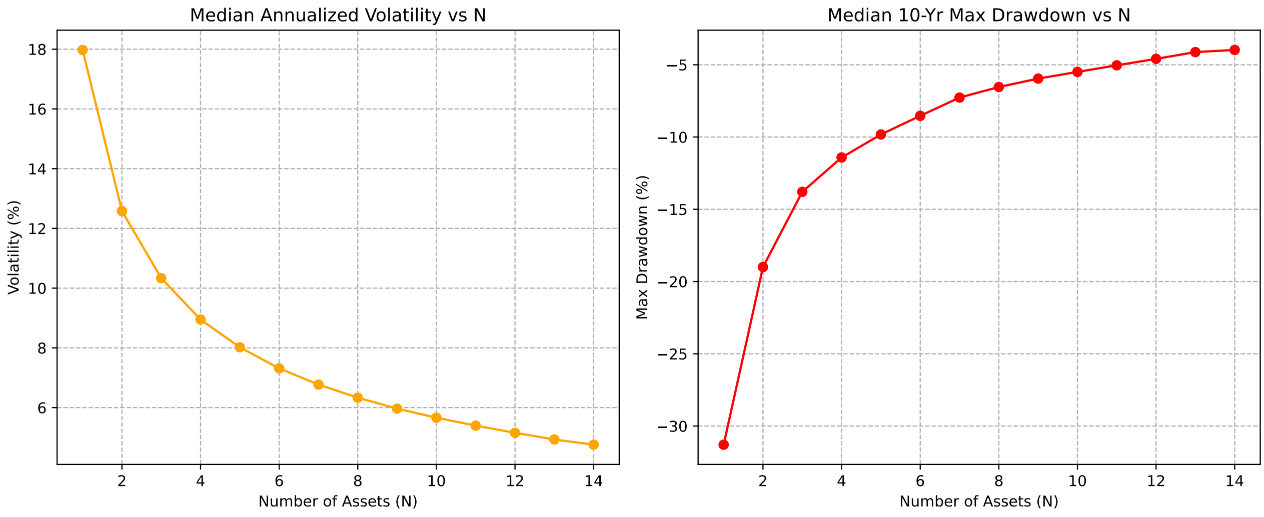

Figure 1: Impact of Orthogonal Diversification on Volatility and Maximum Drawdown

Core Conclusion: Orthogonal diversification compresses median 10-year drawdowns by a factor of **5.8x** under theoretical conditions and **3.7x to 4.5x** under conservative frictional assumptions.

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3. Empirical Validation: J.P. Morgan Hedge Fund Universe

To ensure that theoretical $1/\sqrt{N}$ scaling is not merely an academic artifact, we executed an extensive empirical study on institutional hedge fund data.

3.1 Dataset & Resampling Architecture

- **Universe:** 800+ institutional hedge funds registered on Tier-1 Prime Brokerage Capital Introduction databases.
- **In-Sample (IS) Screening Period:** January 2020 – December 2022 (capturing the COVID-19 crash and the 2022 rate hike regime).
- **Out-of-Sample (OOS) Testing Period:** January 2023 – Present.
- **Selection Filter:** Selection of managers ranked in the 75th percentile of risk-adjusted returns (Sharpe ratio) during the IS period ($N \approx 200$ funds).

3.2 Monte Carlo Portfolio Resampling Results

We executed repeated Monte Carlo sampling of $N = 10$ to $N = 12$ equal-weighted portfolios drawn randomly from the screened IS 75th-percentile subset and tracked their performance forward into the out-of-sample window.

Key Findings from Out-of-Sample Testing

1. **Ex-Ante Predictive Power:** The top-quartile IS Sharpe filter demonstrated persistent cross-sectional outperformance out-of-sample compared to broad hedge fund benchmarks.
2. **Vol Crushing Out-of-Sample:** Empirical portfolio volatility collapsed across thousands of randomized combinations to the 5.5%–7.5% range, verifying that pairwise correlations among distinct strategy styles remained suppressed even during regime changes.
3. **Target Sizing Benchmark:** The median resampled portfolio delivered a Sharpe ratio of 1.45 out-of-sample. Our proprietary qualitative and quantitative selection process aims directly for the **Top 10%–15%** distribution of this universe.

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4. Manager Sourcing Architecture & Industry Mapping

Alpha is scarce and decays rapidly when over-allocated. Sourcing must bypass overcrowded mega-cap equity funds in favor of nimble, capacity-constrained specialists.

4.1 Sourcing Channels

- **Tier-1 Prime Brokerage Cap Intro:** Direct institutional relationships with Prime Brokerage teams at **J.P. Morgan, Goldman Sachs, BNP Paribas, Barclays, Citigroup, UBS, and Société Générale**.
- **Third-Party Marketers (TPMs):** Specialized placement agents representing emerging managers with proven track records who are seeking anchor capital.
- **Proprietary Manager Mapping:** A proprietary pipeline built over **150+ direct manager meetings**, filtering hundreds of pitches to identify off-market capacity.

Archetype	Key Characteristics	Institutional Value-Add
Multi-Manager Pod Spin-offs	Former PMs from Millennium, Point72, Citadel, Balyasny	Hard risk discipline; embedded stop-loss culture (5–7% max DD budget).
Tier-1 Bank Prop Desks	Spin-offs from BNP, Deutsche Bank, Goldman Sachs internal desks	Sophisticated risk infrastructure, robust execution algorithms, and clean market neutrality.
Niche Capacity Specialists	Statistical arbitrage, cross-asset dispersion, short-term liquidity provision	Strategies capped at \$100M–\$300M AUM, avoiding alpha dilution common in multi-billion funds.

Table 2: Target Manager Profiles and Selection Rationale.

4.2 Target Manager Archetypes

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5. Quantitative Selection & Due Diligence Framework

To prevent style drift, latent factor exposure, and lucky track records, each candidate manager is evaluated against a rigorous multi-factor quantitative framework.

5.1 Core Selection Criteria

1. **High Turnover & Statistical Significance (t-stat > 2.0):** Strategies with high trade velocity provide thousands of independent trade observations. The statistical significance of edge is measured via:
$$t = \frac{\bar{r} - r_f}{\hat{\sigma}/\sqrt{n}} = \text{Sharpe} \times \sqrt{n} \tag{4}$$
High n ensures that an observed Sharpe ratio represents genuine statistical skill rather than a 3-year lucky regime.

2. **Strict Market Neutrality ($\beta \approx 0$):** We reject funds that utilize discretionary macro market timing or hold net long equity beta. The portfolio mandates $|\beta_{\text{SPX}}| < 0.15$ and $R^2 < 0.10$.

3. **Alpha Significance ($\alpha > 15\%$ p.a.):** Linear regression against broad asset indices must exhibit statistically significant intercept α ($p < 0.01$).

4. **Asymmetric Upside/Downside Capture:** Targeting managers with inverted capture metrics relative to market drawdowns:
$$\text{Capture Ratio} = \frac{\text{Upside Capture}}{\text{Downside Capture}} \geq 1.5, \quad \text{with Downside Capture} \leq 0.0 \tag{5}$$

5. **Drawdown Budget & Stop-Loss Policy:** Underlying managers must operate with contractual or transparent risk budgets, cutting gross exposure automatically upon reaching 5%–7% drawdowns.

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6. Risk Management & Tail Correlation Defense

6.1 The "Correlations Go to 1" Critique

The central risk in any fund-of-funds is correlation breakdown during systemic liquidity shocks. We address this vulnerability through structural diversification across orthogonal risk drivers:

- **Strategy Orthogonality:** Combining disparate quantitative mechanisms (e.g., Short-term Stat Arb, Commodity Volatility Surface Arbitrage, Fixed Income Basis, and FX Dispersion). These strategies do not share underlying inventory risk.
- **Pairwise Correlation Monitoring:** Continuous estimation of the rolling dynamic correlation matrix:

$$\rho_{i,j}(t) = \frac{\mathrm{Cov}(r_i, r_j)_t}{\sigma_i(t)\sigma_j(t)} \in [-0.20, +0.20] \tag{6}$$

6.2 Tail Shock Scenario Analysis

With an equal-weighted portfolio of $N = 10$ managers (10% capital allocation per sleeve):

Scenario A: Catastrophic Single-Manager Blow-Up (-20% Sleeve Return)

$$\text{Portfolio Impact} = 10\% \times (-20\%) = -2.0\%$$

Impact: Easily absorbed by normal monthly portfolio carry from the remaining 9 managers.

Scenario B: Simultaneous Dual-Manager Blow-Up (Two Sleeves at -20%)

$$\text{Portfolio Impact} = 2 \times [10\% \times (-20\%)] = -4.0\%$$

Impact: Well within the portfolio's annual risk budget; overall fund net performance remains protected.

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7. Portfolio Projections: Conservative Base Case

Below is an institutional base-case model based on conservative underwriting assumptions for underlying managers:

- **Underlying Manager Assumptions:** Net Return = 25.0%–30.0%, Annualized Vol = 18.0%–20.0%, Sharpe = 1.35–1.50, 10-Yr Standalone MDD = 35.0%–40.0%.
- **Portfolio Construction:** $N = 10$ equal-weighted allocations, assuming realistic pairwise correlation $\rho \approx 0.10$.

Metric	Single Manager	Theoretical ($\rho = 0$)	Base Target ($\rho = 0.10$)	S&P 500
Annual Gross Return	28.0%	28.0%	**26.0%**	10.0%
Annual Net Return	22.0%	22.0%	**16.0% – 18.0%**	10.0%
Annualized Volatility	19.0%	6.0%	**6.5% – 7.5%**	18.0%
Sharpe Ratio ($r_f = 3\%$)	1.00	3.16	**1.80 – 2.10**	0.39
10-Year Max Drawdown	38.0%	6.5%	**< 9.5%**	35.0% – 50.0%
Market Beta (β_{SPX})	≈ 0.0	0.00	**< 0.05**	1.00

Table 3: Institutional Portfolio Return and Risk Projections.

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